

```
```sh
sh kops.sh
```
```

```
```sh
export KOPS_STATE_STORE=s3://reyaz-kops-testbkt143.k8s.local
```
```

```
```sh
kops validate cluster --wait 10m
```
```

```
```sh
kubectl get nodes
```
```

PATH Setup (Verbatim)

```
vim .bashrc
export PATH=$PATH:/usr/local/bin/
source .bashrc
```

kops.sh Script (Verbatim)

```
vi kops.sh

#!/bin/bash
aws configure
curl -LO "https://dl.k8s.io/release/$(curl -L -s
https://dl.k8s.io/release/stable.txt)/bin/linux/amd64/kubectl"
wget
https://github.com/kubernetes/kops/releases/download/v1.25.0/k
ops-linux-amd64
chmod +x kops-linux-amd64 kubectl
mv kubectl /usr/local/bin/kubectl
mv kops-linux-amd64 /usr/local/bin/kops

#vim .bashrc
#export PATH=$PATH:/usr/local/bin/
#source .bashrc
```

KOPS Cluster Create (Verbatim)

```
configuration Status=Enabled
export KOPS_STATE_STORE=s3://reyaz-kops-testbkt123.k8s.local
```

```
kops create cluster --name reyaz.k8s.local --zones ap-south-1a
--master-count=1 --master-size t2.medium --node-count=2
--node-size t2.micro
kops update cluster --name reyaz.k8s.local --yes --admin
```

Run the below commands (Verbatim)

```
export KOPS_STATE_STORE=s3://reyaz-kops-testbkt123.k8s.local
kops validate cluster --wait 10m
kops update cluster --name reyaz.k8s.local --yes --admin
kops rolling-update cluster
```

Delete Cluster (Verbatim)

```
# if you want to delete :
kops delete cluster --name reyaz.k8s.local --yes
```

Install HELM (Verbatim)

```
curl -fsSL -o get_helm.sh
https://raw.githubusercontent.com/helm/helm/main/scripts/get-helm-3
chmod 700 get_helm.sh
./get_helm.sh
helm version
```

Install Metric Server (Verbatim)

```
kubectl apply -f
https://github.com/kubernetes-sigs/metrics-server/releases/latest/download/high-availability-1.21+.yaml
```

Add Helm Repositories (Verbatim)

```
# First add helm repositories
helm repo add prometheus-community
https://prometheus-community.github.io/helm-charts
helm repo add grafana https://grafana.github.io/helm-charts
```

Update Helm Chart Repos (Verbatim)

```
helm repo update  
helm repo list
```

Create Prometheus Namespace (Verbatim)

```
kubectl get ns  
kubectl create namespace prometheus  
kubectl get ns
```

Install Prometheus (Verbatim)

```
helm install prometheus prometheus-community/prometheus  
--namespace prometheus --set  
alertmanager.persistentVolume.storageClass="gp2" --set  
server.persistentVolume.storageClass="gp2"
```

```
kubectl get pods -n prometheus  
kubectl get all -n prometheus
```

Create Namespace Grafana (Verbatim)

```
kubectl create namespace grafana
```

Install Grafana (Verbatim)

```
helm install grafana grafana/grafana --namespace grafana --set  
persistence.storageClassName="gp2" --set  
persistence.enabled=true --set adminPassword='Root123456'  
--set service.type=LoadBalancer
```

```
kubectl get pods -n grafana  
kubectl get service -n grafana
```

Access Grafana (Verbatim)

```
Copy the Instance-IP and paste in browser  
username: admin, password = Root123456
```

```
Check SG, nodes and ELB
```